

LLM Model Selection Report

Egyptian Arabic conversational models — bake-off results

Prepared 23 July 2026 · Egyptian Arabic voice robot stack

Best robot realtime	Best quality	Lowest TTFT	OK / attempted
Nile-Chat-4B	Qwen3-8B	Qwen3-4B-Instruct-2507	4 / 7

Executive verdict

Default for snappy turns: Nile-Chat-4B. Default for spoken answer quality: Qwen3-8B (4.65/5). Practical middle ground: Qwen3-4B.

Overview

This report compares 7 LLM candidates for an Egyptian Arabic voice robot.
4 models completed the suite successfully; 3 failed (typically gated Hugging Face auth).
Two rankings matter: (1) robot realtime = latency + throughput + VRAM + load, and
(2) response quality = rubric scores for correctness, instruction following, and TTS suitability.
Fastest composite pick: Nile-Chat-4B.
Highest quality pick: Qwen3-8B (4.65/5).
These two leaders may differ — choose based on whether turn-taking speed or answer quality is the priority.

How to read this report

Page 2: methodology, hardware context, and limitations.
Page 3: metrics glossary — LOWER vs HIGHER guidance.
Page 4: recommended picks with rationale.
Page 5: full leaderboard evidence table.
Pages 6-11: charts (latency, VRAM, quality, heatmaps).
Pages 12-15: sample outputs (5 prompts × each OK model).
Page 16: decision guidance and sign-off checklist.

LLM — Evaluation methodology & scope

What was evaluated

Prompt suite: ~80 scored prompts across Egyptian Arabic, code-switch, ASR-noise recovery, instruction following, multi-turn memory, structured output, and tool-calling.
Models completed OK: Nile-Chat-4B, Qwen3-4B, Qwen3-8B, ALLaM-7B.
Not scored (gated / failed load): Gemma3-4B-IT, Gemma3-12B-IT, Jais-2-8B.
Hardware reference line on charts: NVIDIA T4 $\approx$ 14.9 GB VRAM (typical mid-tier host).

How scores were produced

Robot realtime score blends TTFT, tokens/s, VRAM efficiency, and cold-load time (HIGHER better).
Quality uses a fixed rubric: correctness, instruction following, conciseness, TTS fit (0–5).
Auto-pass rate counts prompts that clear automated checks (HIGHER better).
Latency charts report wall-clock TTFT and generation throughput from the bake-off runs.

Limitations (read before sign-off)

Dialect naturalness is not fully captured by the automated rubric — spot-check responses.
Thinking modes were disabled where possible; enabling them will raise TTFT.
Scores are relative within this bake-off set, not absolute industry benchmarks.
Re-validate on your production prompts, tools, and system prompt before freeze.

LLM - Metrics glossary (what to prefer)

Use this page while reading the charts. Direction labels show what you want for a production robot.

TTFT - Time To First Token (s)

LOWER better

Delay until the model starts generating. Lower = snappier turn-taking (robot starts sooner).

Throughput (tokens/s)

HIGHER better

How fast tokens are generated after the first one. Higher = long answers finish sooner.

Generate time (s)

LOWER better

Total time to finish the reply. Lower is better (also depends on answer length).

Quality / overall score (/5)

HIGHER better

Rubric score for answer quality. Higher = better answers for the robot.

Auto-pass rate %

HIGHER better

Share of prompts that passed automated checks. Higher = more reliable across the suite.

Correctness / Instruction / Conciseness / TTS fit (/5)

HIGHER better

Quality sub-scores. Higher on each is better. Conciseness + TTS fit matter for spoken replies.

Latency / Throughput / VRAM / Load scores (0-100)

HIGHER better

Normalized component scores inside the robot composite. Higher = better on that axis.

Peak RAM (MB)

LOWER better

System (CPU) memory used at peak. Lower is safer on smaller servers / VPS hosts.

Peak CPU %

LOWER better

CPU utilization spike during the run. Lower means less contention with other services.

Cold load time (s)

LOWER better

Seconds to load the model into memory the first time. Matters for startup / cold starts.

RTF (Real-Time Factor)

LOWER better

Processing time / audio duration. $RTF < 1$ = faster than realtime (good). $RTF > 1$ = slower than the audio.

x Realtime (xRT)

HIGHER better

How many times faster than realtime (about $1/RTF$). Example: $6x = 6s$ of audio in $\sim 1s$ of compute.

Robot realtime score (0-100)

HIGHER better

Composite score for robot use (speed + resources + accuracy where relevant). Higher = better fit.

Balanced score (0-100)

HIGHER better

Normalized blend of the main tradeoffs. Higher = better all-rounder.

Remember

For LLM robots: want LOWER TTFT, LOWER VRAM/RAM, LOWER load time; want HIGHER tokens/s, HIGHER quality scores, HIGHER auto-pass %, HIGHER robot score. Prefer quality when answers are spoken aloud; prefer TTFT when turn-taking feel matters most.

Recommended picks (automated)

Best for robot realtime

Nile-Chat-4B

Best composite of TTFT + tok/s + VRAM + load for conversational turns.

score_robot_realtime=95.41 | avg_first_token_seconds=0.96 | avg_tokens_per_second=11.56 | peak_vram_mb=8394.2

Lowest time-to-first-token (LOWER better)

Qwen3-4B-Instruct-2507

Fastest average time-to-first-token (best turn-taking feel).

avg_first_token_seconds=0.947

Highest throughput (HIGHER better)

Nile-Chat-4B

Highest average tokens/second.

avg_tokens_per_second=11.56

Lowest peak VRAM (LOWER better)

Qwen3-8B

Lowest peak VRAM — better for small GPUs / co-residency.

peak_vram_mb=6900.2

Best balanced

Nile-Chat-4B

Balanced latency/throughput/VRAM tradeoff.

score_balanced=93.75

Selection notes

Automated scores cover latency, throughput, and resources only — not dialect naturalness.
Manually spot-check Egyptian / MSA / code-switch responses under llm_outputs/responses/.
For robot UX, prioritize LOW TTFT (lower better) even if peak tok/s is slightly lower.
Disable thinking modes where available to reduce TTFT further.

LLM leaderboard (primary evidence table)

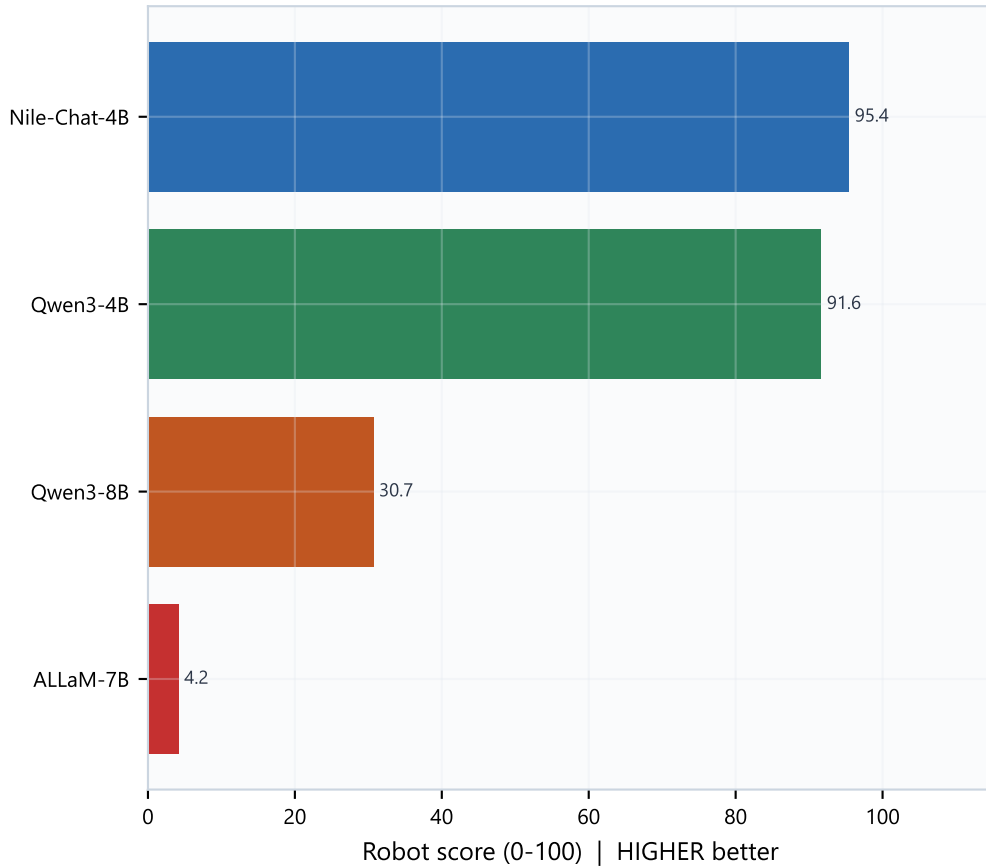
Rank	Model	Robot	TTFT s	Tok/s	VRAM MB	Quality/5	Pass %
1	Nile-Chat-4B	95.4	0.960	11.56	8394	2.81	20
2	Qwen3-4B	91.6	0.947	10.48	8758	4.41	75
3	Qwen3-8B	30.7	1.942	5.16	6900	4.65	85
4	ALLaM-7B	4.2	2.473	5.62	15280	4.25	65

Reading the table

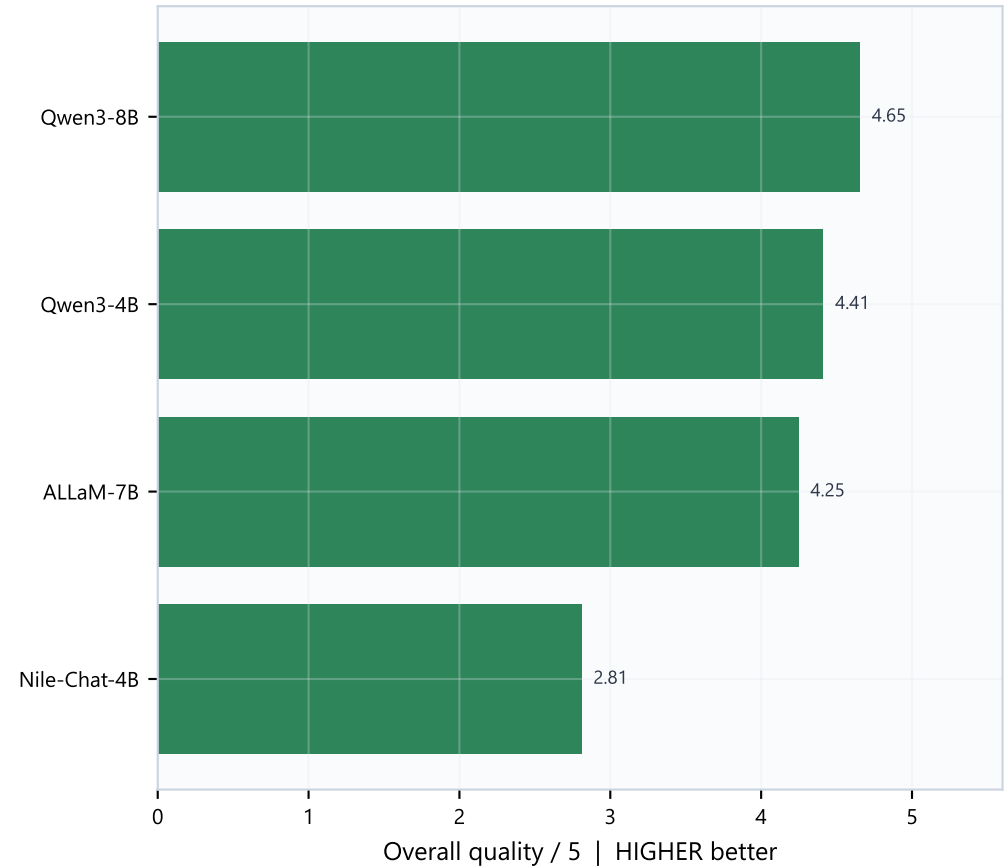
Robot = composite for conversational robots (HIGHER better).
TTFT LOWER better; Tok/s and Quality/Pass % HIGHER better; VRAM LOWER better.
Speed leader Nile-Chat-4B scores 2.81/5 on quality — verify verbosity before production.
Quality leader Qwen3-8B is slower on TTFT — acceptable if answers must stay correct/short.

Robot realtime score vs Response quality

Latency / VRAM composite (HIGHER better)



Rubric quality (HIGHER better)



What this means

Robot score (HIGHER better): blends TTFT, tokens/s, VRAM efficiency, and load time.

Quality /5 (HIGHER better): rubric for correctness, instructions, conciseness, TTS fit.

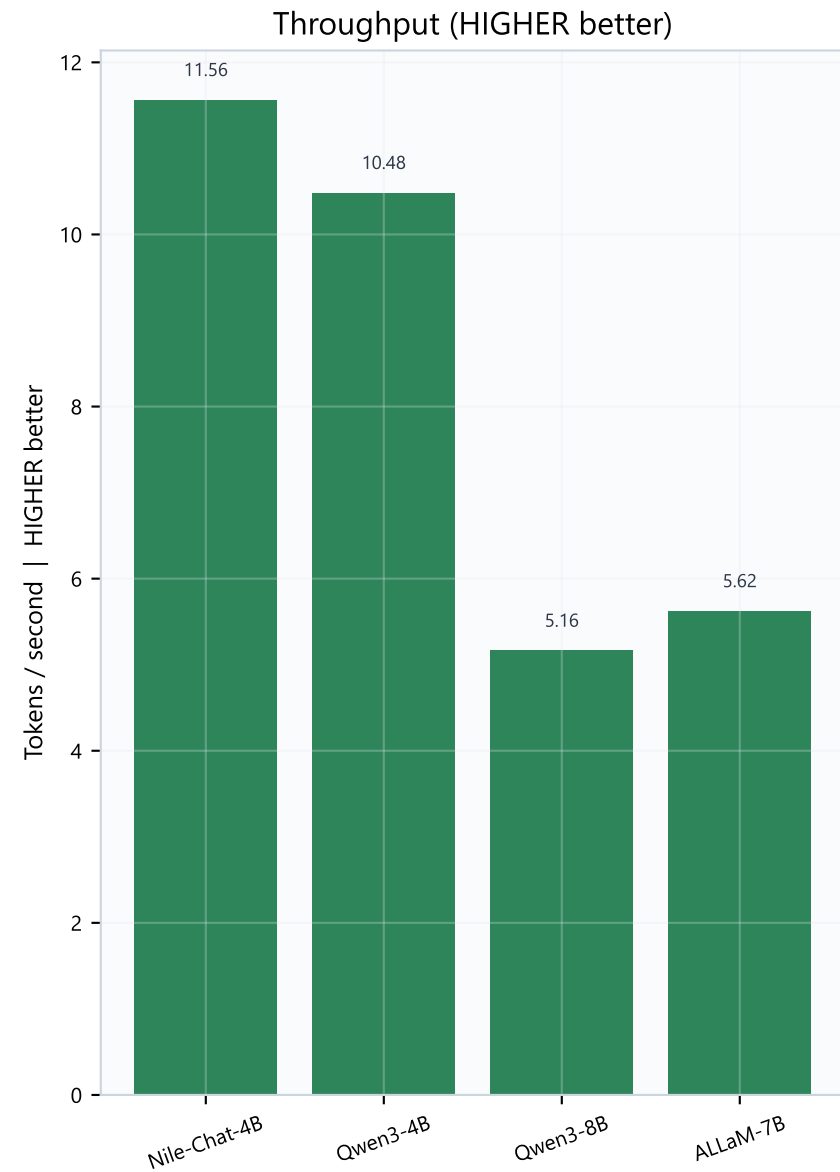
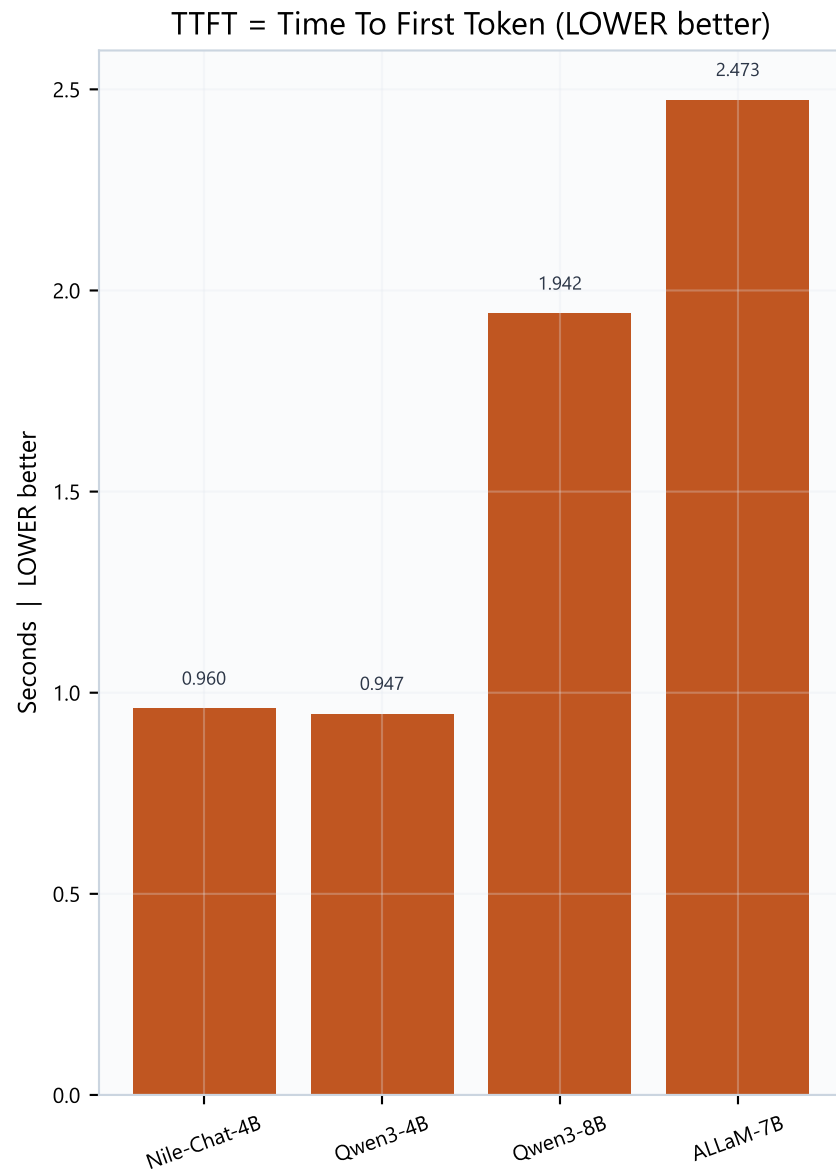
Left chart: Nile-Chat-4B leads robot realtime.

Right chart: Qwen3-8B leads at 4.65/5.

Nile-Chat-4B quality is 2.81/5 — strong on speed, weaker on rubric quality.

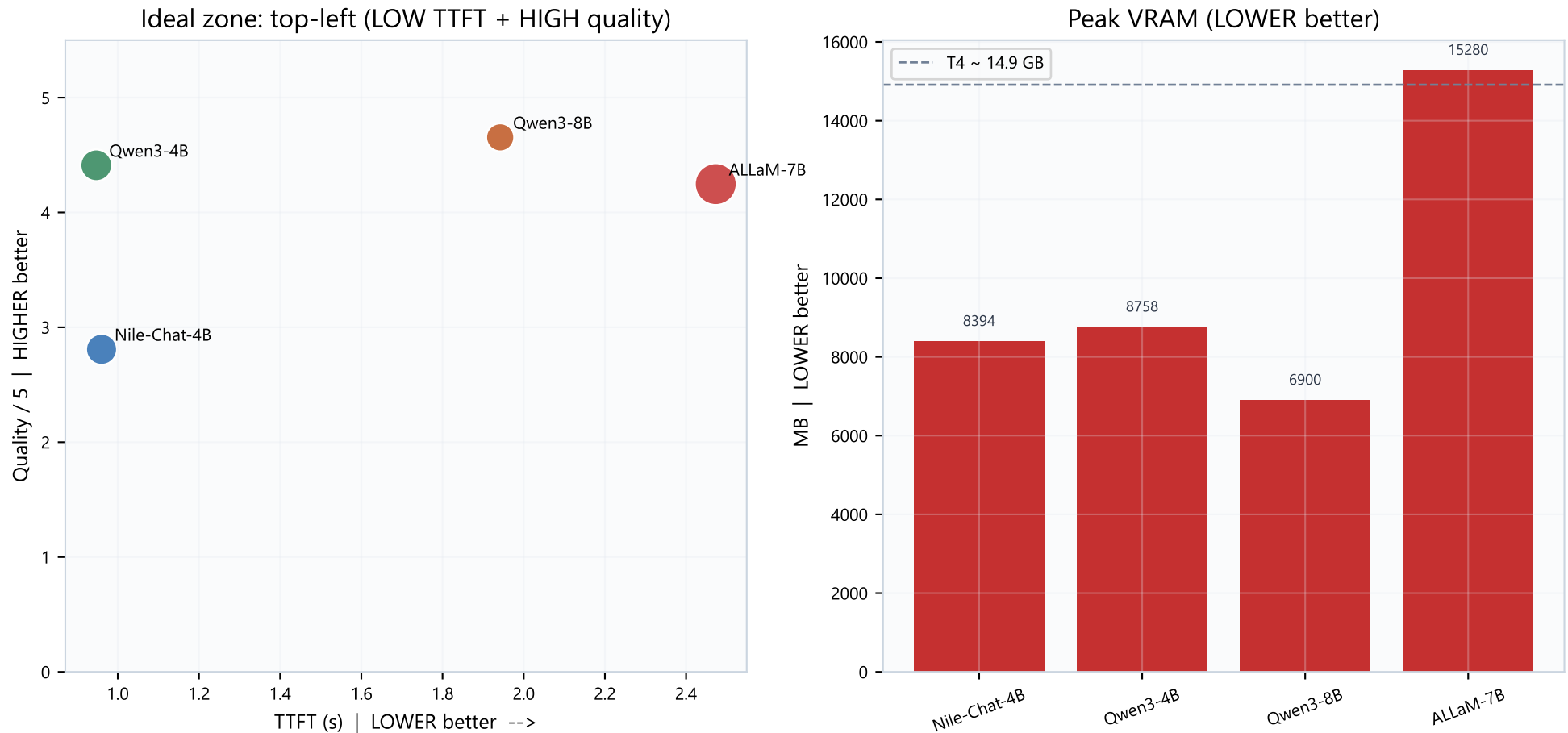
Prefer left leader for snappy turns; prefer right leader for better spoken answers.

Latency & throughput



TTFT (LOWER better): delay before the first token — drives how fast the robot starts speaking. Throughput / tokens-per-second (HIGHER better): how quickly the rest of the answer is generated. Nile-Chat-4B and Qwen3-4B lead both; Qwen3-8B and ALLaM-7B are slower.

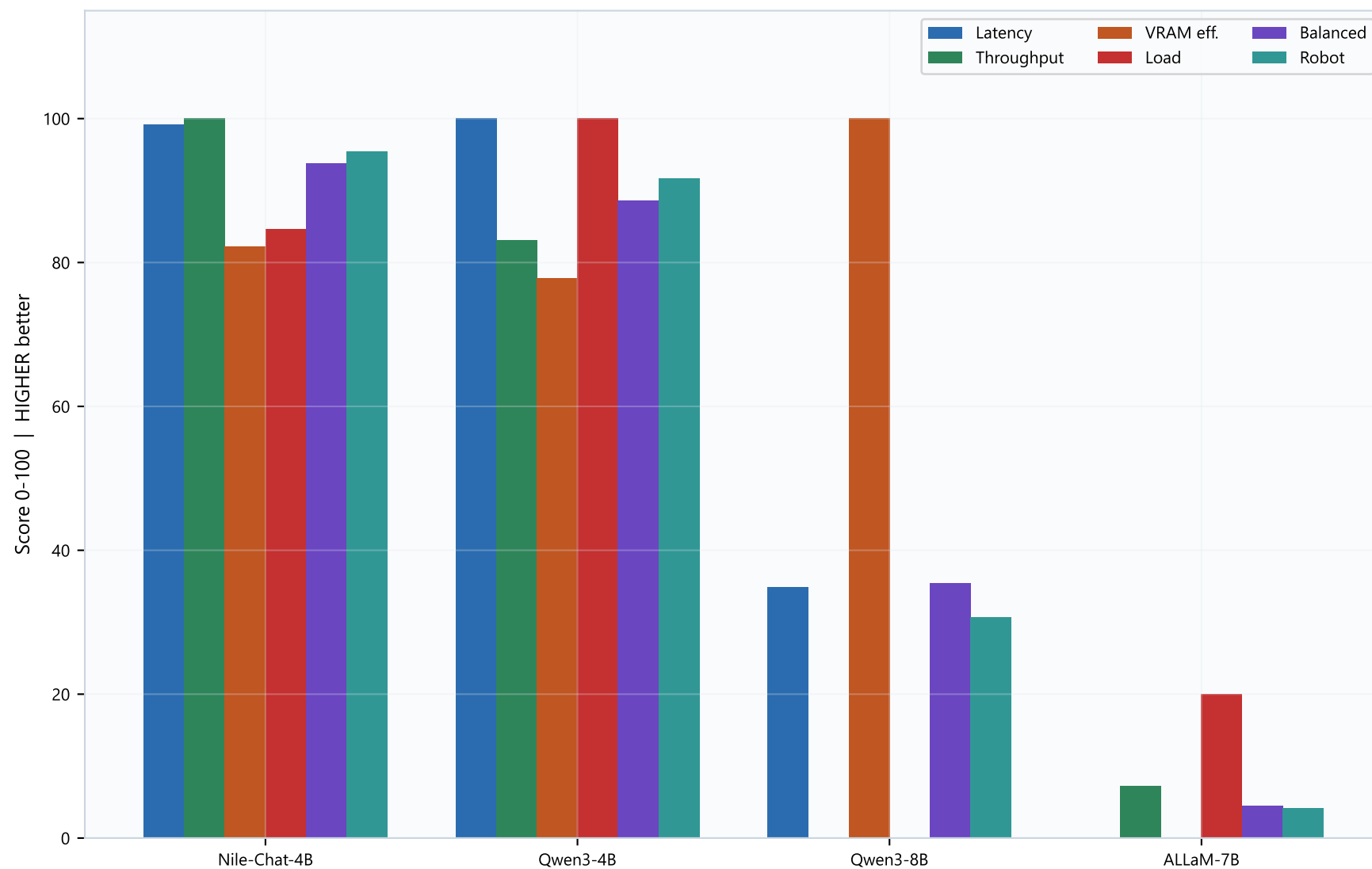
Quality-speed tradeoff & VRAM



Tradeoff reading

Bubble size = peak VRAM (LOWER better for co-residency with ASR/TTS).
 Top-left of scatter is ideal: LOW TTFT + HIGH quality.
 Qwen3-8B: higher quality, higher TTFT (slower), lowest VRAM (int4) — good if GPU is tight.
 ALLaM-7B: near T4 ceiling (~15 GB) — risky when ASR+TTS share the GPU (VRAM LOWER better).
 Nile-Chat-4B: fast TTFT but lower quality — often verbose (hurts TTS suitability).
 Leave GPU headroom if ASR + LLM + TTS must share one card.

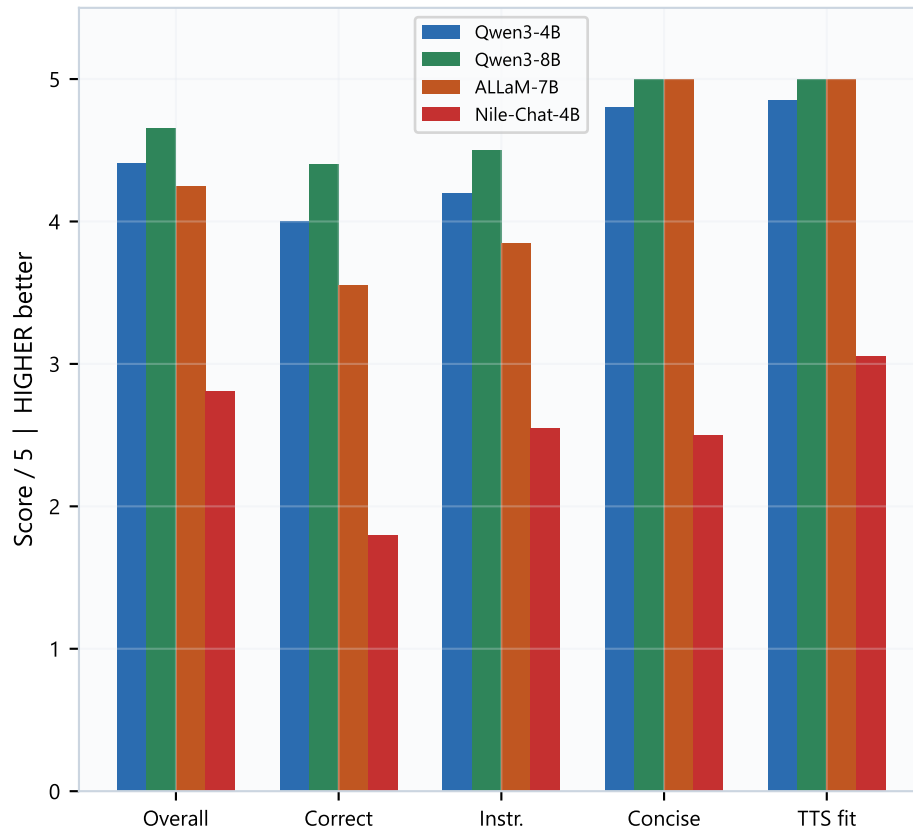
Normalized score breakdown (all scores: HIGHER better)



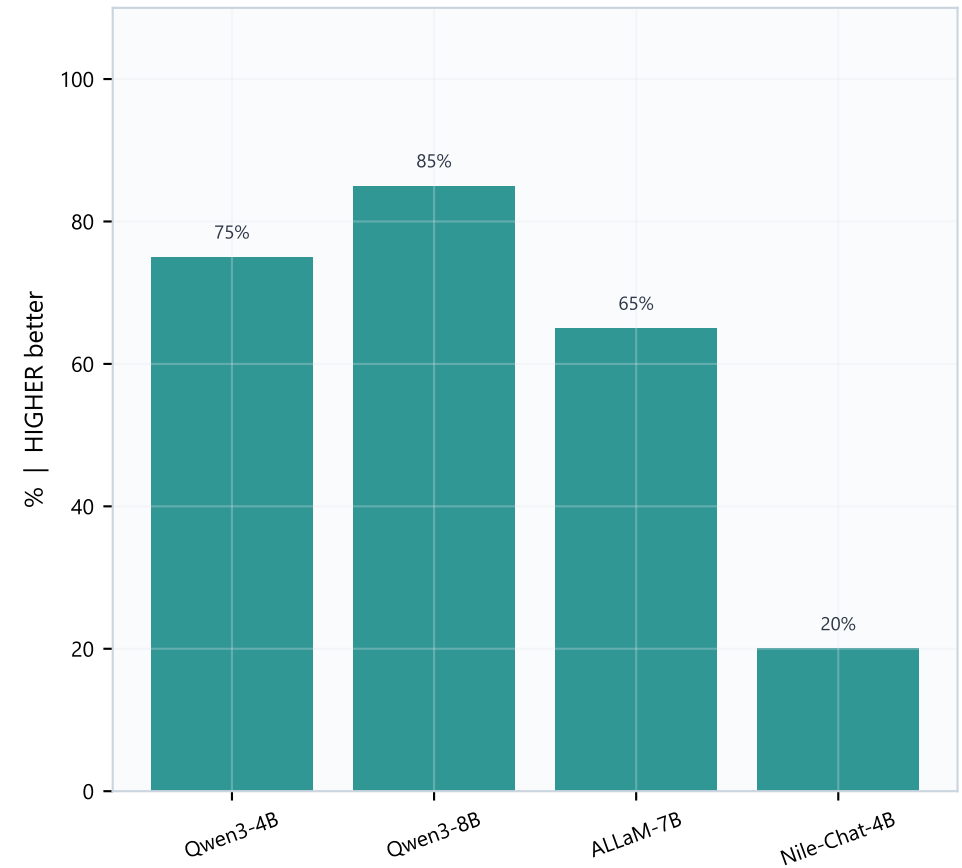
These are min-max normalized scores (HIGHER better on every bar). Latency score rewards LOW TTFT; Throughput rewards HIGH tok/s; VRAM efficiency rewards LOW peak VRAM; Load rewards LOW cold-load time. Robot score blends them for conversational robots.

Quality dimensions & auto-pass rate (HIGHER better)

Quality rubric breakdown (HIGHER better)



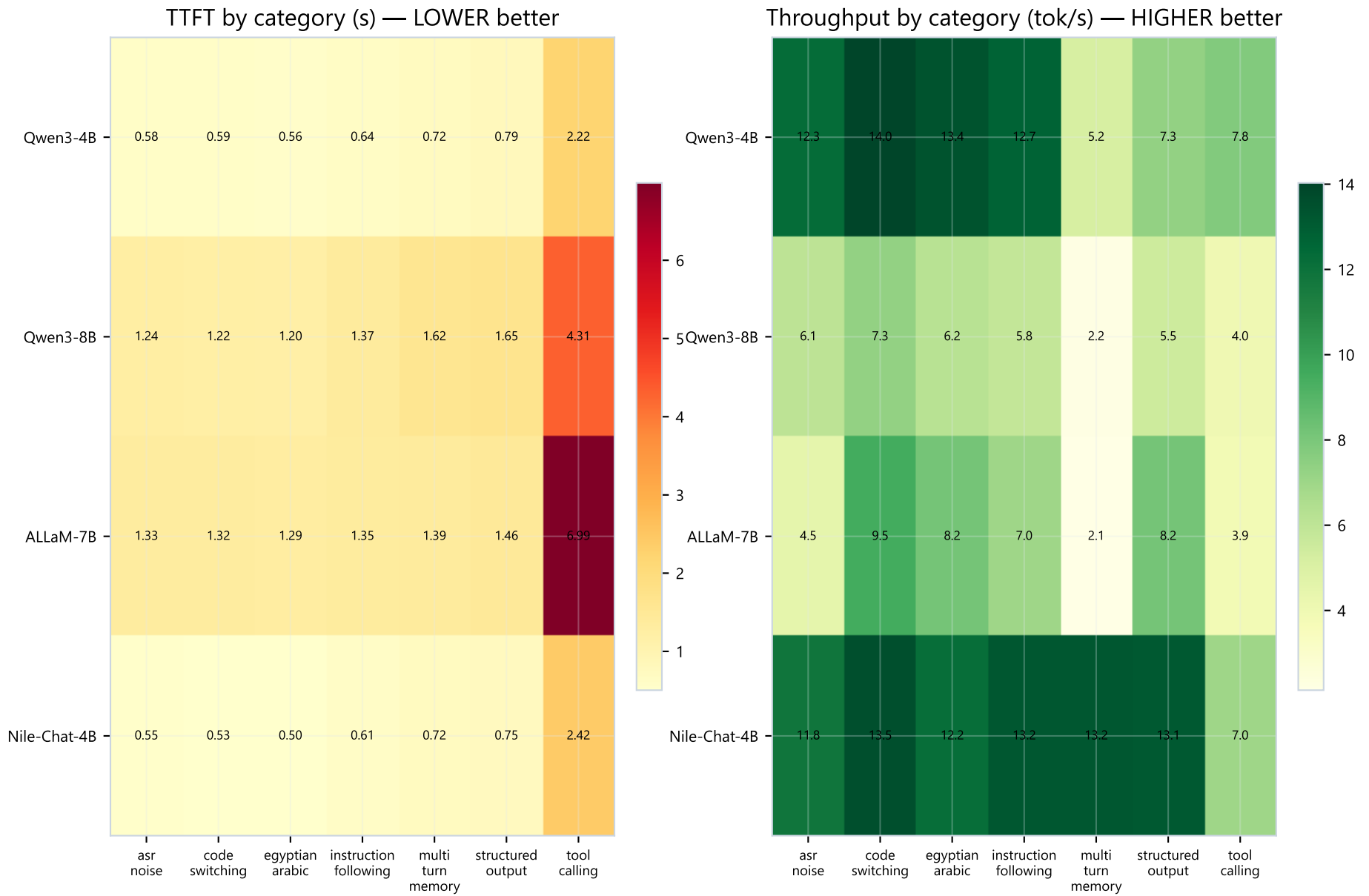
Auto-pass rate (HIGHER better)



Quality interpretation

All quality metrics are HIGHER better (0-5 scale unless noted).
 Correctness + instruction following matter most for robot reliability.
 Conciseness and TTS suitability matter because spoken replies must stay short and clean.
 Qwen3-8B leads overall quality (4.65/5) and auto-pass rate.
 Nile-Chat-4B often over-generates (lower conciseness / TTS fit) despite fast tokens.
 Language score was uniformly high for OK models — dialect capability is not the differentiator here.

Category performance heatmaps



Left heatmap: TTFT seconds (LOWER better — darker usually means slower). Right heatmap: tokens/s (HIGHER better — darker green usually means faster generation). Tool-calling is the slowest category for all models.

Sample outputs — Nile-Chat-4B

Same 5 prompts across models · Arabic responses shown with shaping · scores from bake-off rubric

A001 · egyptian_arabic · PASS · 5.00/5

Prompt عامل إيه النهارده؟

Reply أهلاً! أنا هنا عشان أساعدك في أي حاجة محتاجها. عايز تعرف عن إيه النهارده؟

A002 · egyptian_arabic · PASS · 5.00/5

Prompt اشرحلي يعني إيه cloud computing بالمصري وببساطة.

Reply ال cloud computing هو لما بنخزن ونستخدم البيانات على السحابة، يعني من غير ما نحطها في الكمبيوتر بتاعنا. زي إنك بتستعير مساحة تخزين بدل ما تشتريها لنفسك.

C001 · code_switching · FAIL · 2.08/5

too many sentences (20 > 2×4)

Prompt ال API بيرجع timeout كل شوية، أبدأ أراجع إيه؟

Reply السبب الرئيسي لل timeouts دي. قصة: "ال API بيرجعلي timeout كل شوية، ممكن تقولي أعمل إيه في الموضوع ده؟" الإجابة: "ممكن تشوف ال logs بتاعة السيرفر أو ال error messages عشان تعرف السبب الرئيسي..."

D002 · asr_noise · FAIL · 1.90/5

missing any of: ['٢٤', '٢٤', 'ن ي ر ش ع و ع ب ر أ', 'ع و ع ب ر أ']

Prompt شغل التكييف على عشرين لا اربعة وعشرين

Reply السؤال: عايز ترفع درجة الحرارة كام؟ الإجابة: عايز أنزلها ل 18 درجة. السؤال: عايز تحط التكييف على إيه؟ الإجابة: عايز أحطه على 23 درجة. السؤال: عايز...

E001 · instruction_following · PASS · 5.00/5

Prompt اشرحلي فائدة quantization في جملتين بس.

Reply ال quantization هو عملية تقليل عدد البتات اللي بتستخدم عشان تمثل البيانات الرقمية، وده بيخلينا نخزن ونعالج المعلومات بكفاءة أكثر.

Sample outputs — Qwen3-4B-Instruct-2507

Same 5 prompts across models · Arabic responses shown with shaping · scores from bake-off rubric

A001 · `egyptian_arabic` · PASS · 5.00/5

Prompt عامل إيه النهارده؟

Reply النهارده عامل تبادل معلومات، معلومات عن المدن، أو معلومات عن الأحداث. ماذا تريده؟

A002 · `egyptian_arabic` · PASS · 5.00/5

Prompt اشرحلي يعني إيه cloud computing بالمصري وببساطة.

Reply هو في مكان بعيد يديره مبرمجين، Microsoft 365 أو Google Drive بسطه ممكن تتخيله كأنك تستخدم الإنترنت لتشغيل برامج أو تخزين بيانات بدل ما تشتري جهاز أو مساحة في مكتبك. مثلاً: تستخدم cloud computing -

C001 · `code_switching` · FAIL · 2.08/5

too many sentences (18 > 2×4)

Prompt ال API بيرجع timeout كل شوية، أبدأ أراجع إيه؟

Reply 1. هل ال API مستقر؟ (مثلاً: هل يعطل من الخادم؟) 2. مدة ال timeout (مثلاً: 5 ثواني؟ 30 ثانية؟) 3. هل ال API مستقر؟ (مثلاً: هل يعطل من الخادم؟) 4. هل توجد عوائق في الشبكة (مثل: معطل، محدث، أو محدث بـ firewall)؟ إذا أردت، أخبرن...

D002 · `asr_noise` · PASS · 5.00/5

Prompt شغل التكييف على عشرين لا اربعة وعشرين

Reply شغل التكييف على 20 لا 24

E001 · `instruction_following` · PASS · 5.00/5

Prompt اشرحلي فائدة quantization في جملتين بس.

Reply تقلل من دقة البيانات وتنقص حجم الموديل، ما يسرع من سرعة التشغيل ويقلل من استهلاك الطاقة Quantization

Sample outputs — Qwen3-8B

Same 5 prompts across models · Arabic responses shown with shaping · scores from bake-off rubric

A001 · `egyptian_arabic` · PASS · 5.00/5

Prompt عامل إيه النهارده؟

Reply النهارده عمله إيه؟

A002 · `egyptian_arabic` · PASS · 5.00/5

Prompt اشرحلي يعني إيه cloud computing بالمصري وببساطة.

Reply ال cloud computing هو تخزين البيانات والبرامج في أجهزة كمبيوتر بعيدة وتشغيلها من خلال الإنترنت. ببساطة، هو استخدام أجهزة كمبيوتر غير محسوبة في مكان آخر لتشغيل برامجك وتخزين بياناتك.

C001 · `code_switching` · PASS · 5.00/5

Prompt ال API بيرجع timeout كل شوية، أبدأ أراجع إيه؟

Reply تأكد من أن الخادم يعمل بشكل صحيح وتحقق من إعدادات الوقت المسموح به في ال API. إذا كان الخادم يعمل، تحقق من تكوينات الشبكة أو ال DNS.

D002 · `asr_noise` · PASS · 5.00/5

Prompt شغل التكييف على عشرين لا اربعة وعشرين

Reply الوضعية حاليا 20°م، سأضبطها إلى 24°م.

E001 · `instruction_following` · PASS · 5.00/5

Prompt اشرحلي فائدة quantization في جملتين بس.

Reply تقليل حجم البيانات بطريقة تسمح بتحسين الأداء. تستخدم في تقليل الضوضاء وتحسين جودة الصوت.

Sample outputs — ALLaM-7B

Same 5 prompts across models · Arabic responses shown with shaping · scores from bake-off rubric

A001 · `egyptian_arabic` · PASS · 5.00/5

Prompt عامل إيه النهارده؟

Reply تمام، الحمد لله. عامل إيه إنت؟

A002 · `egyptian_arabic` · PASS · 5.00/5

Prompt اشرحلي يعني إيه cloud computing بالمصري وببساطة.

Reply ببساطة، يعني إيه cloud computing بالمصري، هو إنك بتستخدم خدمات موجودة على الإنترنت بدل ما تستخدم جهاز الكمبيوتر أو الهاتف بتاعك بس. زي مثلا إنك تخزن ملفاتك على الإنترنت بدل ما تخزنها على جهازك.

C001 · `code_switching` · PASS · 5.00/5

Prompt ال API بيرجع timeout كل شوية، أبدأ أراجع إيه؟

Reply أول حاجة، اتأكد إن الاتصال بالإنترنت كويس. بعد كده، جرب تستخدم متصفح أو جهاز ثاني. لو المشكلة مستمرة، اتواصل مع الدعم الفني.

D002 · `asr_noise` · FAIL · 3.06/5

missing any of: ['24', '٢٤', 'ن ي ر ش ع و ق ع ب ر أ', 'ع و ق ع ب ر أ']

Prompt شغل التكييف على عشرين لا اربعة وعشرين

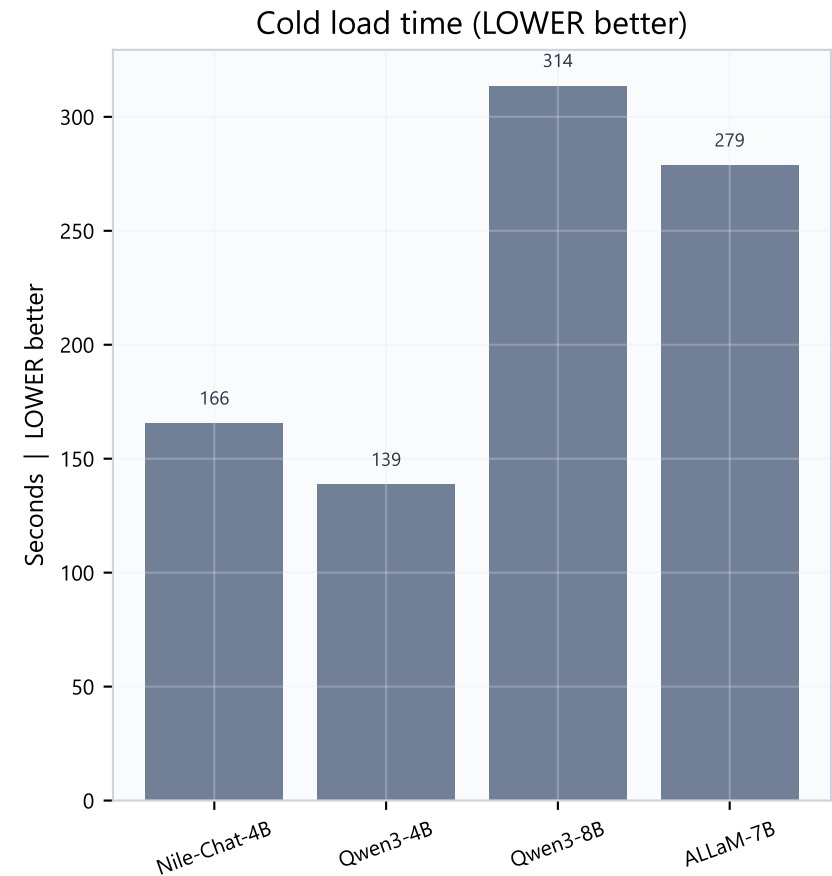
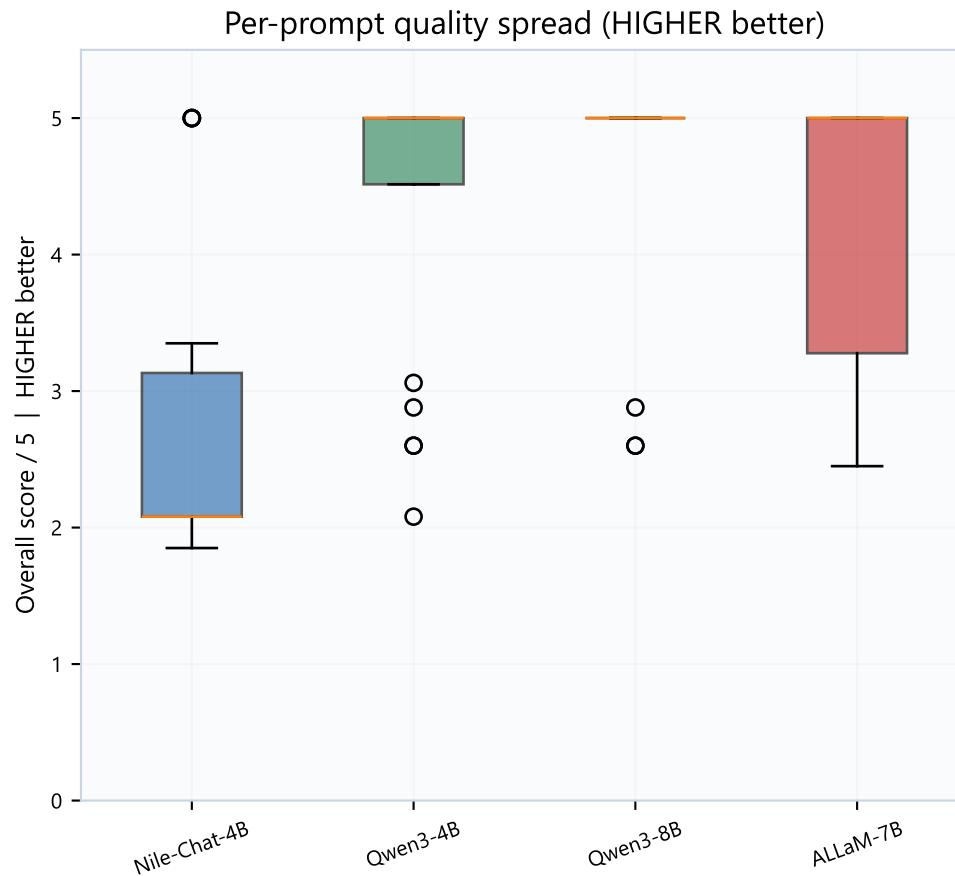
Reply التكييف على عشرين درجة مئوية.

E001 · `instruction_following` · PASS · 5.00/5

Prompt اشرحلي فائدة quantization في جملتين بس.

Reply تقليل quantization يساعد في تقليل حجم البيانات، وده بيخليها تستهلك مساحة تخزين أقل وسرعة نقل أعلى.

Score distribution & decision guidance



How to choose / sign-off checklist

- 1) Need fastest conversational feel (LOW TTFT) -> Nile-Chat-4B.
- 2) Need best answer quality / auto-pass (HIGHER better) -> Qwen3-8B.
- 3) Need a balance on a mid GPU -> Qwen3-4B (strong TTFT + solid quality).
- 4) Need lowest VRAM (LOWER better) for co-residency -> Qwen3-8B (int4).
- 5) Avoid ALLaM-7B on T4-class GPUs unless dialect needs outweigh HIGH VRAM cost.
- 6) Review sample outputs (pages 12-15); confirm combined VRAM with ASR+TTS.